

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz and exam.

Problem 1. (10 points) Find the following limits using the L'Hospital's Rule.

$$(a) \lim_{x \rightarrow +\infty} \frac{\ln x}{\sqrt{x}}$$

$$= \lim_{x \rightarrow +\infty} \frac{\frac{d}{dx} [\ln x]}{\frac{d}{dx} [\sqrt{x}]} = \lim_{x \rightarrow +\infty} \frac{\frac{1}{x}}{\frac{1}{2\sqrt{x}}} = \lim_{x \rightarrow +\infty} \frac{2\sqrt{x}}{x} = \lim_{x \rightarrow +\infty} \frac{2}{\sqrt{x}} = 0$$

$$(b) \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{d}{dx} [1 - \cos x]}{\frac{d}{dx} [x^2]} = \lim_{x \rightarrow 0} \frac{\sin x}{2x} = \lim_{x \rightarrow 0} \frac{\frac{d}{dx} [\sin x]}{\frac{d}{dx} [2x]} = \lim_{x \rightarrow 0} \frac{\cos x}{2}$$

$$= \frac{1}{2}$$

$$(c) \lim_{x \rightarrow 0^+} x^x \quad (\text{Hint: apply natural logarithm})$$

$$y = x^x$$

$$\ln y = x \ln x$$

$$\lim_{x \rightarrow 0^+} x \ln x = \lim_{x \rightarrow 0^+} \frac{\ln x}{\frac{1}{x}} = \lim_{x \rightarrow 0^+} \frac{\frac{d}{dx} [\ln x]}{\frac{d}{dx} [\frac{1}{x}]} = \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{-\frac{1}{x^2}} = \lim_{x \rightarrow 0^+} -x = 0$$

thus

$$\lim_{x \rightarrow 0^+} x^x = \lim_{x \rightarrow 0^+} e^{\ln y} = e^{\lim_{x \rightarrow 0^+} \ln y} = e^{\lim_{x \rightarrow 0^+} x \ln x}$$

$$= e^0$$

$$= 1$$